Agents, mobility and multimedia information

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This paper describes the design philosophy and implementation of a system which manages the location, retrieval and processing of multimedia information for mobile customers. The system uses intelligent agents in all aspects of management and allocation of service components to perform the most appropriate translation and movement of information through the network.

The agents use an open market model to provide the services. The strategy of the management agents is to stimulate demand to use their services, which is offset by quality-of-service factors, leading to balanced utilisation of the network. Agents also act as proxies for the user to take into account personal preferences.

1. Introduction

Multimedia services present a telecommunications provider with many interesting options when compared to traditional voice telephony. The latter is based around a single, standard offering over which universal and common services are provided. It is questionable whether such an approach is either desirable or obtainable with multimedia. A plethora of sources and displays are available, varying from phones and video cameras that plug into telephone networks, to top-end workstations capable of displaying pseudo-3D virtual reality graphics. When users want to send multimedia documents they will have to give a great deal of thought to what equipment the receiver has available to display the document, and to the ability of the network to deliver it at a reasonable transmission rate. This complexity will inhibit the use of such services. Simply, users want to send documents and leave the 'network' to make intelligent choices about where and how to process, translate and display them.

These problems are particularly acute for the mobile user. Mobile equipment rarely has the display or processing power of 'fixed' items due to the need for low-power consumption and robust build quality. Thus the mobile user will only have access to lower processing power terminal facilities. Additionally, the mobile customer is normally attached to the network via a wireless tail which normally has less bandwidth than the fixed network. In the future the situation will almost certainly become more complex with users able to access information over an enormous number of different networks each of which will have different advantages and limitations.

The provision of intelligent management systems that select the optimum format of reception and the best network for the purpose and cost will be essential to give customers the necessary flexibility, and network operators a competitive edge. The network should be capable of taking information and delivering it to the user in the most appropriate form, depending on the equipment in the vicinity and the user's personal preferences. It is this problem that has been addressed using intelligent agents to provide the flexibility for a solution that is:

- robust — in that the network solution should be resilient to or be able to route round component failures or provide alternative solutions,
- scalable — so that the solution will work for local networks, but also be appropriate for large corporate networks or at a national or international level,
- flexible — so that new components that become available can be introduced into the system and used where appropriate with the minimum of intervention or manual re-configuring of the system.

The rest of this paper goes on to describe the rationale behind the agent approach used in the Multimedia Information Interchange (MII) system, then outlines the architecture and different types of agents that are involved in this system. At this point the approach is compared with alternative methods of achieving the same goal. The final section gives examples of the kind of services that can be achieved from the office-based prototype that has been in operation for the last few months.

2. Design philosophy and architecture

For the purposes of this paper, an intelligent agent is an autonomous computer program that continuously learns about its environment, adjusts its behaviour appropriately and negotiates with other agents to buy and sell resources needed for the overall system to undertake some task. Autonomy, learning and negotiation are the basic attributes of this agent definition [1].

A number of key ideas about what agents are and how agent systems should function drove the design of the agent model and architecture. As can be seen from the list below, the three major criteria in designing an agent-based network and service management system [2] are scalability, flexibility and robustness:

- each agent manages one resource, thus allowing for flexibility and robustness — a resource is viewed as either a component part of a service, such as a basic translation service, up to a whole network, which may be managed by an agent or hierarchy of agents,
- it should be possible to remove any agent from the system so that it still operates effectively but possibly in a reduced capacity,
- there should be a minimum messaging overhead — it is essential that, when there are some ten million users, the inter-agent messages do not swamp the system,
- agents should form hierarchies (scalability).

Fundamentally, there are three different problems the agents will need to tackle in this multi-network environment — the location of information, the allocation of network resources and the tracking of people and equipment. All have different characteristics, although only the latter two are actually implemented here, the proposed agent model can be extended to cover all three.

To best describe the architecture of the system it is first necessary to revisit briefly the service that the system is performing. Basically it has to deliver a multimedia document to a mobile user in the most acceptable form making best use of the resources it has available. This problem is depicted in Fig 1. If the receiver of the information is at a compatible workstation the multimedia document can be transmitted directly. If they only have access to a PDA (personal digital assistant), it will be unlikely to have the display capabilities and so the document must be decomposed, and perhaps only the text transmitted. The most extreme case is also shown where the only access to the person is via a phone. In this case a précis of the document must be made and that précis translated to speech to be delivered over the phone.

To be able to perform this task effectively the system needs to know:

- the preferences of the recipient,
- the capabilities of the equipment near the recipient,
- the translation facilities that are available in the network.

One of the requirements was to ensure that the customer knew the cost of delivery of the message before it was sent, as some of the translation services may not be free. The delivery of a message is therefore split into two phases — a high-level pass to determine the possible delivery formats

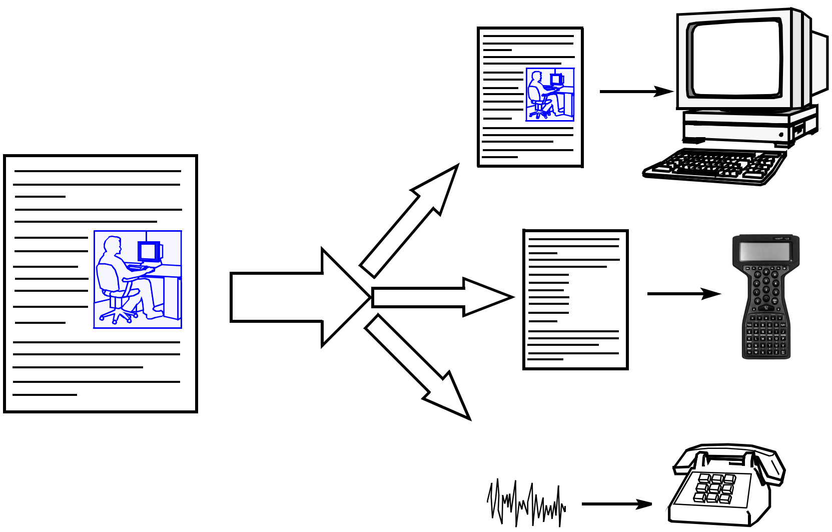

Fig 1 The ubiquitous service problem.

and cost, and whether it is possible to deliver it at all, given the cost and time constraints that may be imposed by the user. Then, once agreed, a second pass to actually implement the service in terms of routeing the message through the appropriate translation services to allow the message to be delivered in the correct form.

2.1 The first phase

Figure 2 shows the agent communication involved in the management agent negotiation. The icons represent different types of agents, the lines between these icons represents the communication between the agents (the thicker the line the greater the communication). In Fig 2, negotiation between a geography database, network agent and two customer agents is shown. During this negotiation the agents calculate the document formats and cost of delivery by using statistical knowledge of the network resources. This is analogous with the management structure of a company where a customer would request a service from the manager of a group. This manager knows approximately what resources are available to him, how long the job will take and how much it will cost. Using this information the manager provides a quote for the work.

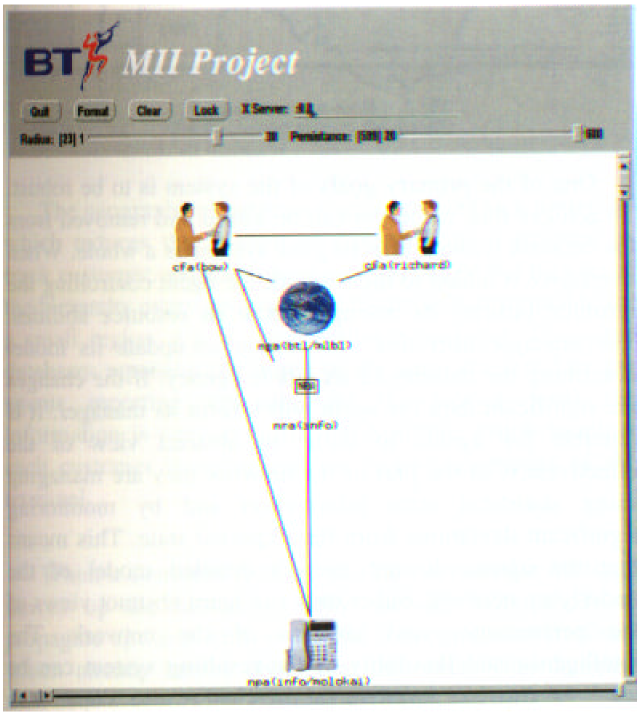

Fig 2 Management agent negotiation.

2.2 The second phase

Each translation service is controlled by resource agents which combine dynamically to provide any type of translation service that is necessary. The resource agents are involved in the second stage of the negotiation. This can be seen in Fig 3, where the resource agents (for the link, service and terminal resources) can be seen in a negotiation to determine exactly which resources are needed in the service and in what order they should occur. Returning to the analogy of the management structure, this phase of the negotiation is the same as the employees of the manager working out how they will do the work required. Each multiple translation service requested is built on the fly in this way.

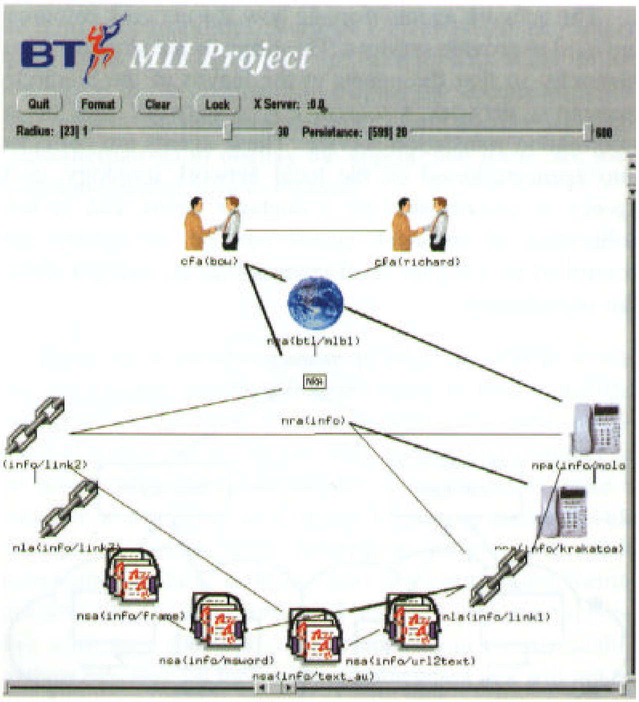

Fig 3 Resource agent negotiation.

2.3 The agents involved

A number of different types of agent have been constructed to perform these various resource management tasks. There are agents managing the user interfaces to the network which are known as customer facing agents (CFAs). Other types of agent manage the network services at a high level and are known as NRAs, and a third group manage the low-level network resources such as translation services and links (NSAs and NLAs). To keep track of the users and current location of equipment a hierarchical geographic database is used.

Customer-facing agents

All customers subscribing to the network have a personalised CFA. This agent provides a single point of contact and is responsible for learning the customer's profile allowing the best to be obtained from the network.

This agent negotiates with the geography and the network agents to determine how the service can be provided. As the network agents may return with more than one possibility, an offer is sent to the customer describing the service which best matches the request. If the customer accepts the offer then the customer-facing agent buys the service in from the network, otherwise the customer-facing agent may provide another offer if more choices exist.

Network agents

The network agents manage how the network resources are used to provide services. These agents are structured in a hierarchy so that the agents at the leaves of the hierarchy manage a network resource, e.g. a network link or an information translation service. These agents are collected into agencies based on the local network topology; each agency is co-ordinated by a manager agent, and in turn collections of manager agents can be an agency and controlled by a higher level manager agent — Fig 4 shows this organisation.

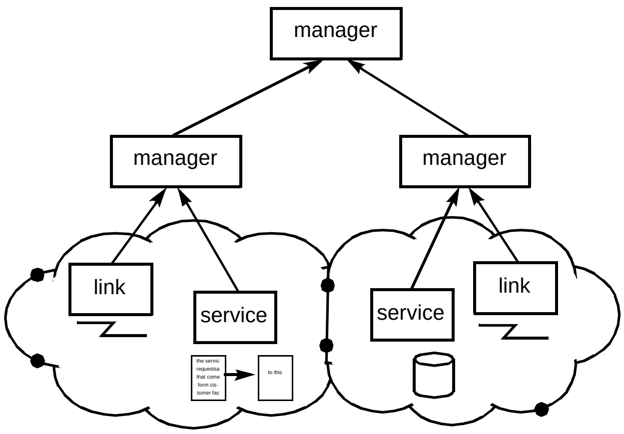

Fig 4 Network resource agents.

The service requests that come from customer-facing agents are processed starting at the top of the hierarchy and filtering downwards. The manager agents compare the requirements of the service with their network model. If the agents' network could be used in the service then the agent will enter a bid. This bid is used to construct the service offer for the first phase of the service as discussed earlier. If the agents' bid is accepted then the manager must negotiate with the agents in its agency to provide the resources required to fulfil its bid (the second phase). If the managers fail to provide the service or the service provisioning fails, this is indicated up the hierarchy and this failure is used by the learning mechanisms to alter the network model in the managers to reduce the chance that unsuccessful services will be contracted in the future.

The peer-to-peer bidding process of the resource manager agents used to link together the necessary service components performs a distributed search through the resources taking account of the availability, cost and abilities of the different resources. A representation of this search can be seen in Fig 5. The icons represent different media formats and agents. In this example, the service needs to translate a Microsoft Word document (on the left) to speech output (on the right). Terminal agents (represented by the telephone icons on both sides) move the information into and out of the network. A service agent can be seen converting the Word document into text. This search graph is only partially complete. A number of 'gaps' can be seen where the search has yet to be completed. It can be seen that a gap exists between the text document and the speech file. As the negotiation between the agents continues from this point, an agent managing a text-to-speech resource will be consulted and will be able to fill this gap.

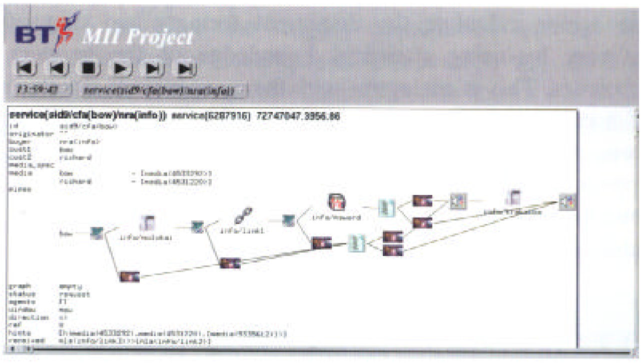

Fig 5 Resource search graph.

One of the primary goals of the system is to be robust. To achieve this, resources can be added and removed from the network without affecting the system as a whole. When a resource is added to the network the agent controlling the resource informs its manager about its resource abilities. The manager uses this information to update its model describing the abilities of its whole agency. If the changes are significant then the agent will inform its manager. It is possible for agents to build an abstract view of the effectiveness of the part of the network they are managing using statistical state information and by monitoring significant deviations from the expected state. This means that the agents do not need a detailed model of the underlying network; rather they can learn abstract views of the performance and abilities of the network. The intelligence and flexibility of the resulting system can be used to provide a universal mechanism to cope with updating, network failure and service management.

2.4 *Geography databases*

The geographic database needs to provide a flexible, fast search that enables the network to track people and equipment and their relationship. A number of distributed database models exist that have been exploited or analysed.

The Home Location Register-Visiting Location Register is used in current mobile networks and the REMUS architecture [3] is more closely linked with the system used here. The design philosophy used was to build a distributed database using the same tools as used in the network domain and so keep the system design simple. As will be seen, this gives a particularly robust and flexible database.

To provide the best use of the terminal equipment available to the customer, the network needs to know the geographical relationship between the customer and the terminals (see Fig 6). The geography databases are responsible for maintaining and using information about these relationships.

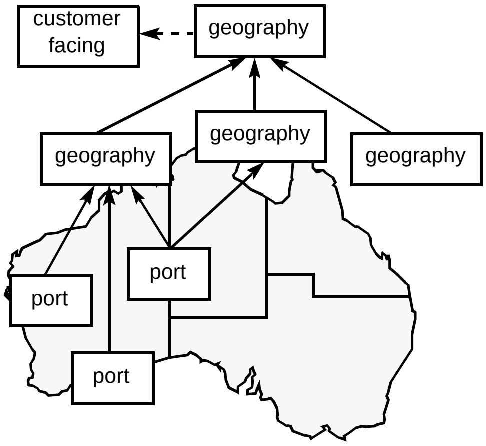

Fig 6 Geographical databases.

The geographical databases are organised in a hierarchy which reduces the amount of communication required to track customer and equipment movement. At the bottom of the hierarchy every geographical database is responsible for a small region, e.g. a floor within a building. The other databases represent the union of the areas managed by the agents reporting upwards. At the lowest level the information is very detailed, for example which terminal each customer is using and the exact capabilities of the terminal.

When a customer logs on to a new terminal a message moves up the hierarchy. This message only needs to propagate upwards until it reaches a geographical database which already knows about that customer. This database then adds a new pointer to the sub-hierarchy indicating the customer has logged on in a new place. Log-off operations are handled in a similar way. This system reduces the communications overhead when customers are moving around different areas of the network because most of the communication occurs locally to the customer. When a service is requested via the customer-facing agent the requirements from the customer include the media type of the information being transmitted and constraints on the delivery media (if any). The geographical databases use the location and terminal information to recommend terminals near the customer for this service.

3. Discussion

From what has been described, the system is quite complex with interactions between different types of agents to perform what may appear to be quite a simple delivery service. After all, the WWW can deliver multimedia documents to the desktop in a fairly straightforward way and, in theory, mime-compliant mail readers can also offer a document, delivered in the most appropriate form depending on the terminal capabilities. Various Web browsers use helper applications to do this which are fired up depending on the mime-type of the document to display the document, while mail readers can interpret multipart mime documents (multipart/alternative) to display the appropriate form. So, why is such a complex system needed?

3.1 Why not the WWW model?

There are a number of reasons why the WWW model was not thought suitable. The first thing to note is that the document source, be it Postscript, plain text, html, etc, is made available ‘as is’ on the Web server. The provider of the information cannot reasonably be expected to make it available in a number of different formats to please a wide audience (although some benevolent people have made documents available both as (say) Microsoft Word format and html). If a user wants to look at one of these documents, they must have the appropriate helper application available on their machine. In the most common cases this will not be a problem, but it is unlikely that everyone will have every helper application available to be able to read the documents. The majority of the Unix community for example will be unable to read any Microsoft Word document that may be made available.

With the MII approach, this is not a problem. If the appropriate helper application is not available, the document is translated, on-the-fly, to a format that is most suitable for the end user. With the translation facilities being network-based, there is no computational overhead at the destination end, and the service provider can make available the latest versions of the translation software, ensuring a highly effective service.

Bandwidth may be an issue; some Web sites are notoriously slow, or the delay might be on the receiving end. The user may only have available a low-bandwidth link, such as a modem to connect to the network. In these cases, the user has no option but to wait for the whole download to see if the document is worthwhile. With MII this could be taken into account by the user’s customer-facing agent which would place a limit on the amount of data to be sent at any one time, and any documents to be delivered would be routed through some summarisation service [4] to ensure maximum content for minimum bandwidth.

3.2 Why not the intelligent mail-reader approach?

Mime-compliant mail readers are able to take mail of the form multipart/alternative and deliver the document in the most appropriate form for the terminal capabilities available. The argument against this is simply that the document needs to be made available in all the different formats that are possible. For example a document written in Microsoft Word would ideally need to be made available in the following formats:

- as is,
- rich text format (in case you do not have the same version of Microsoft Word),
- plain text format,
- summarised plain text.

This approach is not thought to be practical, as it would place much more effort on the part of the document provider to try to second-guess the formatting capabilities of the reader of the document. Additionally, this method also probably doubles or triples the amount of data needed to be sent across the network, which again may be an issue if one of the links is of low bandwidth.

3.3 Related work

The IBM Intelligent Communications Service [5] claims to provide services which are very similar to those in MII. Routeing and translation allows the customers of this service to communicate over a variety of networks and terminals. Differences between the systems exist on how the service to be delivered is determined. With the IBM system, a set of customer-specified rules [6] determine the routeing and translation required by the Intelligent Communications Service, whereas in MII the translations and networks to be used are determined dynamically based on the availability of translation services in the network and terminals near to the customer. The current status of IBM's Intelligent Communications Service is unknown.

3.4 Do we need agents at all?

The problem tackled here was to provide a very flexible method of delivering information, making appropriate choices of delivery format. The advantages that agents bring in this application are the following:

- a very robust approach to the management of the delivery service, covering failure of elements and allowing dynamic re-routeing,
- flexible configuration management — new translation services are added locally, and the management agents pass on the availability of these new services as necessary,
- a personalised approach from the user's viewpoint allowing information to be delivered in the most appropriate form, learned over a period of time.

4. Example services

A demonstrator system has been operational in an office environment for a number of months, allowing a range of the services and facilities, that the agent system provides, to be demonstrated. The prototype runs over a LAN, and integrates a number of Sun workstations, including a portable. A number of media-translation services have been provided to the system such as text-to-speech [7], Microsoft Word-to-text, FrameMaker-to-text, etc.

4.1 Document translation

The first example service depicts one customer sending a Microsoft Word document to another customer who can only be contacted using a mobile phone. Service requests from the customer consist of a number of attributes which describe the service required (Fig 7). The request contains a contact address for the person to whom the document should be sent, the document type, when the document needs to be received, the cost of the document and who should be billed. The template which the customer is required to fill in is shown in Fig 7. In this example, a Microsoft Word document is being sent to Richard which needs to be received as soon as possible. The icons at the bottom of the form indicate that a Microsoft Word document is being sent by the customer and the format in which the document is received is unknown.

Once the service request has been received by the agents, a high-level negotiation occurs between the customer-facing agents and the network agents, making use of the geographical database. The agents realise collectively that it is possible for the document to be translated into a text document then into a speech file and read out over the mobile phone. This results in a service offer being sent to the customer as shown in Fig 8. The icons on this form show that the document will be converted into speech for delivery to the customer at the other end of the phone. The customer then has a choice to accept or reject the offer made to the network. If the offer is accepted then the network agents enter the second stage of negotiation that sets up the necessary resources for the service. The information is then

Fig 7 Customer request.

Fig 8 Network offer.

received by the network, processed as necessary and then delivered to the receiving customer.

4.2 *Document redirection*

The redirection of services to a more appropriate terminal near the customer has also been demonstrated. For example, a VRML (virtual reality modelling language) document can be sent to a customer who is using a workstation with an unsuitable graphics card. The agents exploit the geographical knowledge about the relationship of the terminals to determine that a high specification workstation is nearby. Both customers are then informed where the document is to arrive. Where possible the network will strive to achieve the highest accuracy of transfer at the specified cost, rather than downgrading the message type simply because the recipient is not currently on a suitable machine. This decision is made by the manager agents during the first pass of the negotiation.

In Fig 9, the graph constructed during the first phase of this service is shown. On the left it can be seen that a VRML document is being sent into the network. On the right all the possible target document types for the service are shown. It can be seen in the figure that some of these types are repeated; this is because the receiving customer is near two terminals, one a high-power workstation which can cope with many different format types, the other a simple PDA with text and limited graphics abilities.

This exploitation of the communication and computing environment around a user allows the system to create a virtual communications shell wherever the customer moves. The network carries out this function continuously without input from the user — a key feature of the ubiquitous communications system vision of the future.

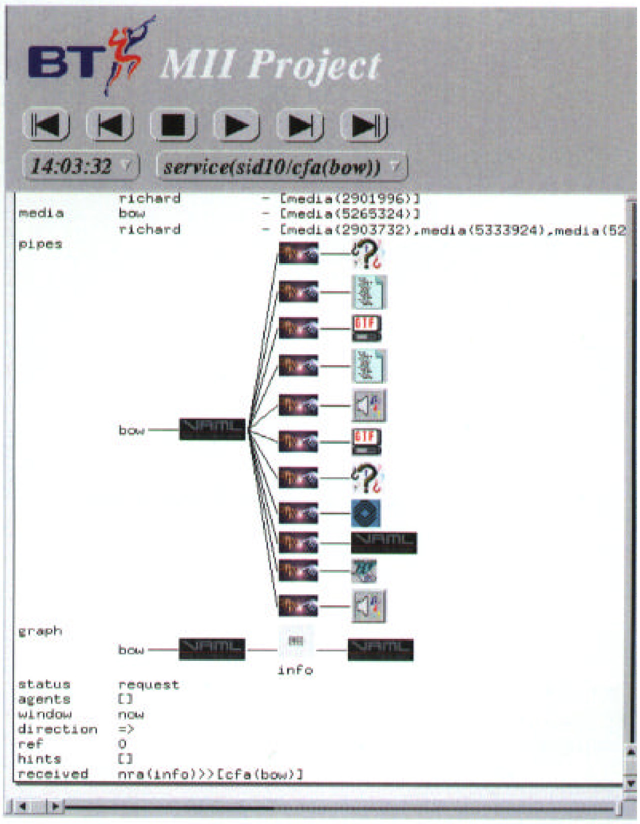

Fig 9 Negotiation during a redirection service.

5. Conclusions

It has been shown that communities of intelligent agents can be used for effective management of a flexible document delivery service. The ability of agents that manage services being able to bid to supply services provides a simple but powerful mechanism to give flexibility in the use of service components and enables performance management to be handled in a dynamic manner.

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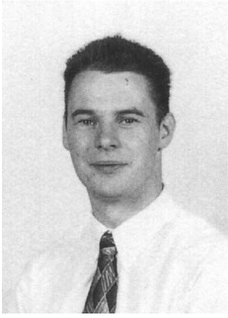

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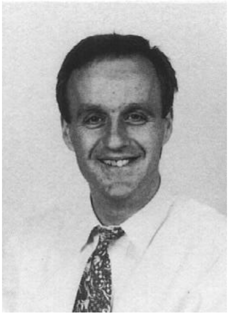

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